POLY PMNA · DIPLOMA CURRICULUM REV2026

Course 5009 | Semester 5 | Project | Shared handbook

Major Project - Phase I

Build the approved foundation for a major project: problem definition, literature or background study, requirements, feasibility, design, planning and review evidence.

Course code	**5009**
Revision	**REV2026**
Semester	**Semester 5**
Type	**Project**
Catalogue scope	**42 catalogue programme assignments**

COURSE ORIENTATION

How to use this handbook

This page is a structured learning aid for the shared catalogue record. Study the concept, complete the applied activity, retain the evidence, and compare the result with the latest approved programme instructions. Where the official syllabus differs, the official record takes precedence.

Source and verification note: The course identity, code, semester and subject type are taken from the tracked POLY PMNA REV2026 catalogue, which indexes the official SITTTR scheme. The official course-content reference is SITTTR course 5009; the scheme index is SITTTR REV2026 diploma syllabus. The direct subject-content page was not machine-readable or returned an unavailable-file response during preparation, so module headings and explanatory teaching material on this page are an original study-handbook structure, not claimed verbatim SITTTR wording. Confirm current hours, marks, credits, outcomes and assessment against the latest official programme record before examination use.

Learning outcomes

- Explain the central concepts and vocabulary of the subject using clear technical language.
- Apply a controlled project workflow to a diploma-level problem with scope, milestones and acceptance evidence.
- Create a proposal, plan, design or final evidence pack that a supervisor can review.
- Identify assumptions, limitations and common errors, then improve the work using feedback.

Shared-record context

This code is assigned to 42 catalogue programme assignments in the tracked catalogue. The page therefore focuses on common learning practice and avoids claiming programme-specific technical details that were not verified.

Study rule: Do not treat the author-created examples as official examination questions or official mark distribution.

STUDY MAP

Modules and evidence

Author-created study map; verify official hours and marks			
Module	**Heading**	**Purpose**	**Evidence to retain**
1	Project problem and outcomes	Define a substantial, programme-relevant problem and convert it into measurable project outcomes.	Prepare a problem statement and an outcome table with measurable acceptance indicators.
2	Background study and requirements	Use reliable sources and stakeholder evidence to establish what the project must do.	Create a background-study matrix and a first requirements specification.
3	Feasibility, risk and resources	Test whether the project can be completed safely and credibly with available time, skills, tools and budget.	Complete a probability-impact risk register and a feasibility decision record.
4	Concept design and project plan	Select and justify a concept, then plan its execution and evidence collection.	Submit a concept-review pack with diagram, alternatives, schedule and review questions.
5	Phase-I review and handover	Present a defensible foundation so Phase II can begin without hidden assumptions or uncontrolled scope.	Use a review checklist and record every decision, action owner and due date.

MODULE 1

Project problem and outcomes

Purpose-led study | Apply + reflect

Define a substantial, programme-relevant problem and convert it into measurable project outcomes.

Core topics–

- Need, context and stakeholders
- Problem statement and objectives
- Scope, exclusions and constraints
- Expected technical and social value

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Prepare a problem statement and an outcome table with measurable acceptance indicators.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 2

Background study and requirements

Purpose-led study | Apply + reflect

Use reliable sources and stakeholder evidence to establish what the project must do.

Core topics−

- Literature and related-solution review
- Source evaluation and citation
- Functional and non-functional requirements
- User stories, use cases and traceability

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Create a background-study matrix and a first requirements specification.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 3

Feasibility, risk and resources

Purpose-led study | Apply + reflect

Test whether the project can be completed safely and credibly with available time, skills, tools and budget.

Core topics−

- Technical, economic and operational feasibility
- Risk register and mitigation
- Resource and procurement plan
- Ethics, safety and data protection

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Complete a probability-impact risk register and a feasibility decision record.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.

3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 4

Concept design and project plan

Purpose-led study

Apply + reflect

Select and justify a concept, then plan its execution and evidence collection.

Core topics−

- Architecture, process or system diagram
- Alternative comparison and decision criteria
- Work breakdown, milestones and responsibilities
- Verification strategy and review gates

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Submit a concept-review pack with diagram, alternatives, schedule and review questions.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 5

Phase-I review and handover

Purpose-led study | Apply + reflect

Present a defensible foundation so Phase II can begin without hidden assumptions or uncontrolled scope.

Core topics−

- Proposal and review presentation
- Approved requirements and design baseline
- Open issues and decisions
- Phase-II implementation plan

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Use a review checklist and record every decision, action owner and due date.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

ASSESSMENT PREPARATION

Evidence, revision and quality gates

Before submission or examination

1. Confirm the current SITTTR/SBTE title, hours, credits, outcomes and assessment.
2. Define the problem or prompt and state assumptions.
3. Show the method, calculation, algorithm, project record or communication structure.
4. Attach test, source, supervisor, workplace or reflection evidence as applicable.
5. Review clarity, safety, ethics, citations, originality and accessibility.

Revision checklist

- Can I define the main terms without copying?
- Can I apply the method to a new situation?
- Can I explain one common error and its correction?
- Can I show evidence for each claimed outcome?
- Can I state the limitations and the next improvement?

```
Evidence record
Subject: 5009 — Major Project - Phase I
Task or question:
Assumption or requirement:
Method / design / procedure:
Result or artefact:
Test / source / supervisor evidence:
Reflection and next action:
```

SELF-CHECK AND EXAMINATION PRACTICE

Question bank

Recall question

Define two important terms from Major Project - Phase I and distinguish them.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Explain question

Explain why the method in this handbook matters for a diploma student.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Apply question

Apply one module method to a realistic technical, project or workplace situation.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Evaluate question

Identify a weak approach, state the evidence needed, and propose a defensible improvement.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Create question

Design a small artefact, plan, test, report section or presentation that demonstrates the subject outcomes.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

REFERENCES AND GLOSSARY

Reference trail

1. Official SITTTR REV2026 diploma syllabus catalogue.
2. Official SITTTR course-content reference for code 5009.
3. POLY PMNA Study Hub, for the current revision index and related lesson pages.

Glossary

Terms used in this handbook

Term	Working meaning
Outcome	A capability the learner should demonstrate, not just a topic read.
Evidence	A record, artefact, result, source, test or review that supports a claim.
Acceptance criterion	A condition used to decide whether a requirement or deliverable is satisfactory.
Reflection	A brief evidence-based account of what worked, what did not and what should change.

Prepared as a POLY PMNA study handbook. Always follow the latest official SITTTR/SBTE instructions for the programme and examination session.